



DE - Projekt: Digital Engineering of Process Engineering Applications

Kickoff Presentation

MachViz

High-Fidelity & Adaptive Visualization
of Compressible Nozzle Flow

Group 3

Meet the Team

CEE

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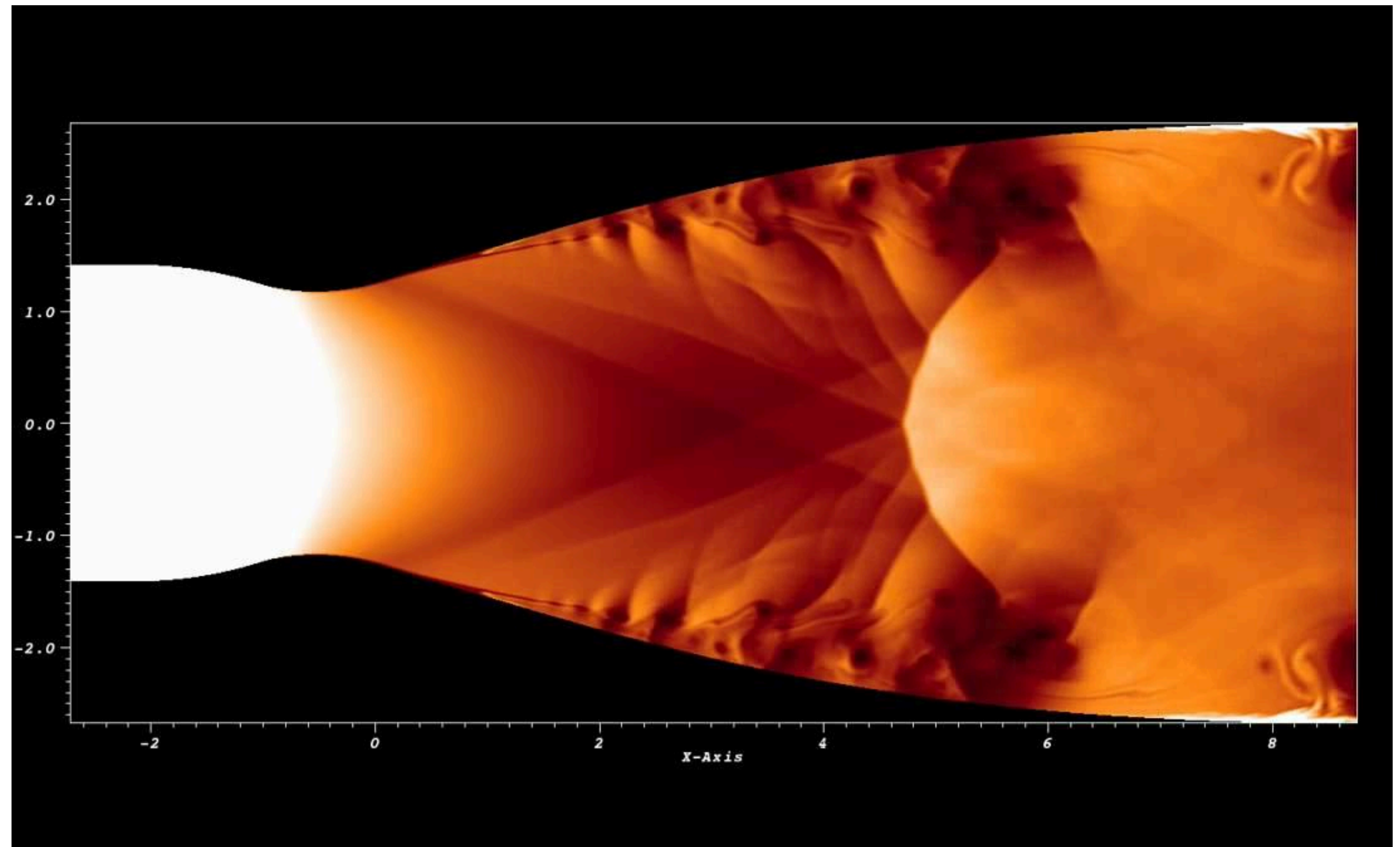
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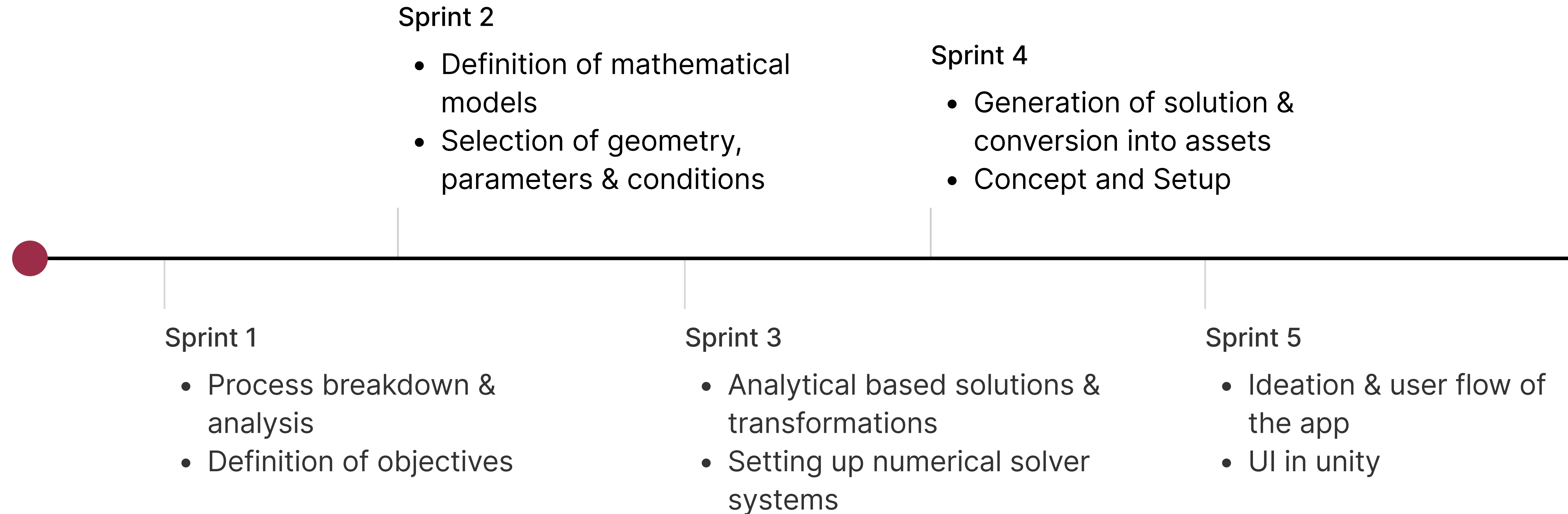
Topic

FLOW THROUGH CONVERGENCE DIVERGENCE NOZZLE

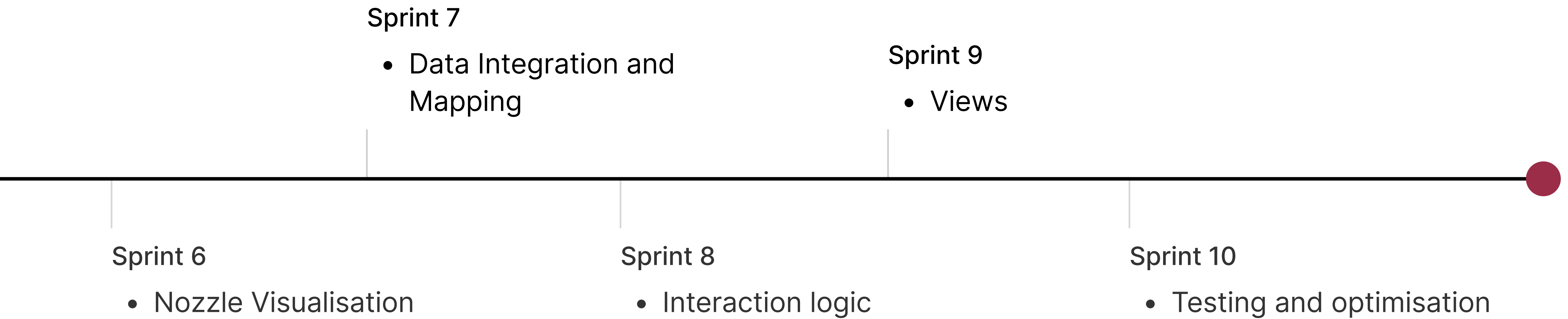
- Development of a simple solver to attain velocity, temperature & pressure fields using a combination of analytical & numerical methods.
- Implementation of the solver in steady 2D inviscid & viscous flows at subsonic, supersonic & shockwave flows.
- Visualization of the using unity assets or imported blender assets.
- Creation of an interactive interface, where user can change different parameters to see the change in flow properties.



Timeline



Timeline



Work Distribution

Jaideep

Model creation & solver
development for **subsonic** nozzle
flow

Theresa

Model creation & solver
development for **supersonic** nozzle
flow

Shashank

Model creation & solver
development for nozzle **shockwaves**

Hemanth

Visualization & Interaction

Rakshan

Graphics & Rendering

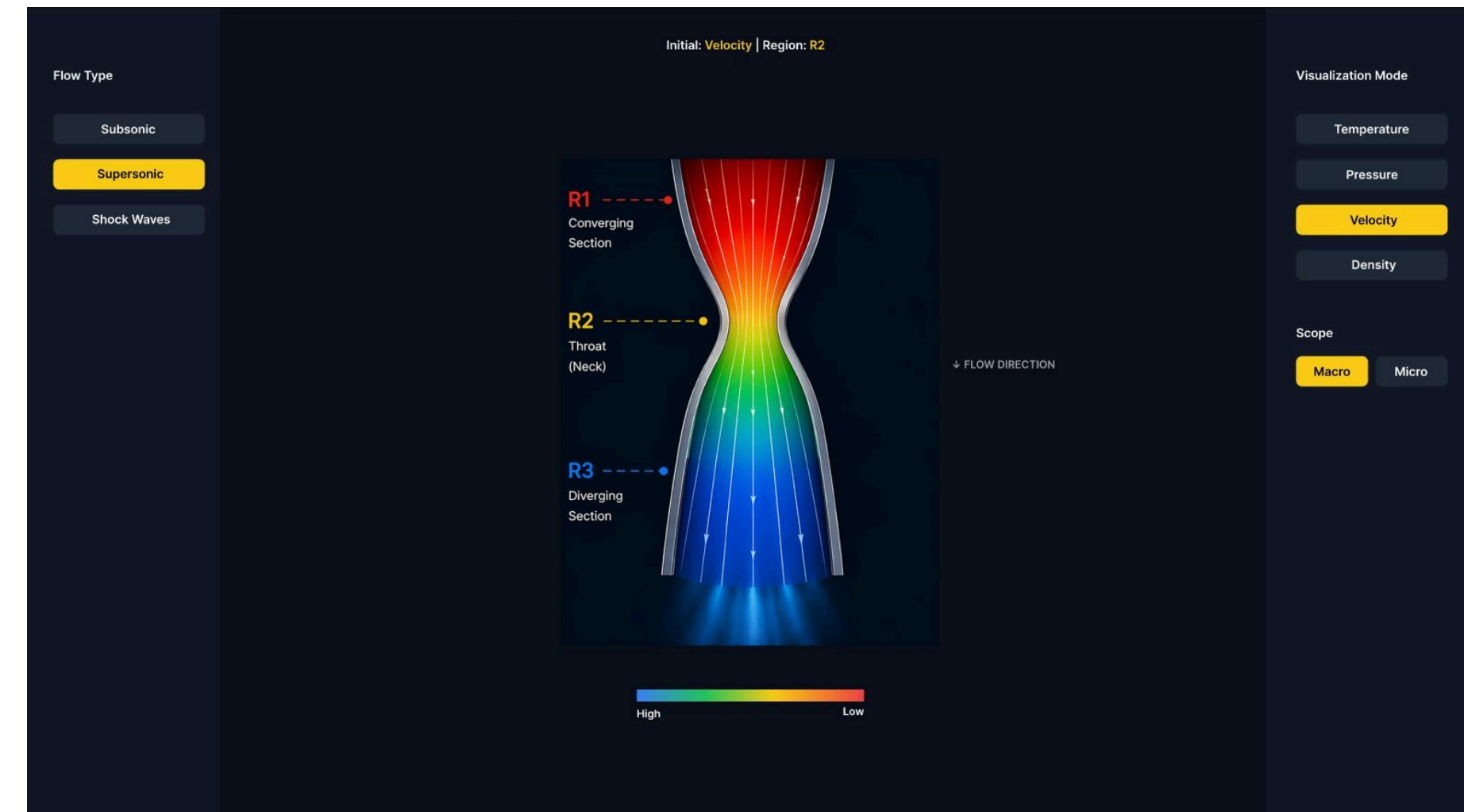
Mantasha

Data pipeline & Logic

VISUALIZATION SCHEME

Interface includes:

- High-Fidelity screens displaying the flow fields inside the nozzle
- Interactive features for the user to navigate between different flow conditions



The screenshot shows a 'Simulation Setup' dialog box with the title 'Select initial conditions to start visualization'. It contains two columns: 'Initial Variable' and 'Region'. The 'Initial Variable' column has buttons for 'Temperature', 'Pressure', 'Velocity' (selected), and 'Density'. The 'Region' column has buttons for 'R1', 'R2' (selected), and 'R3'. A 'Start' button is located at the bottom right of the dialog box.

Flow Type

- Subsonic
- Supersonic
- Shock Waves

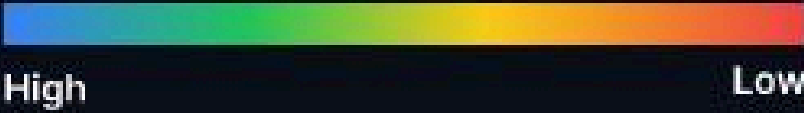
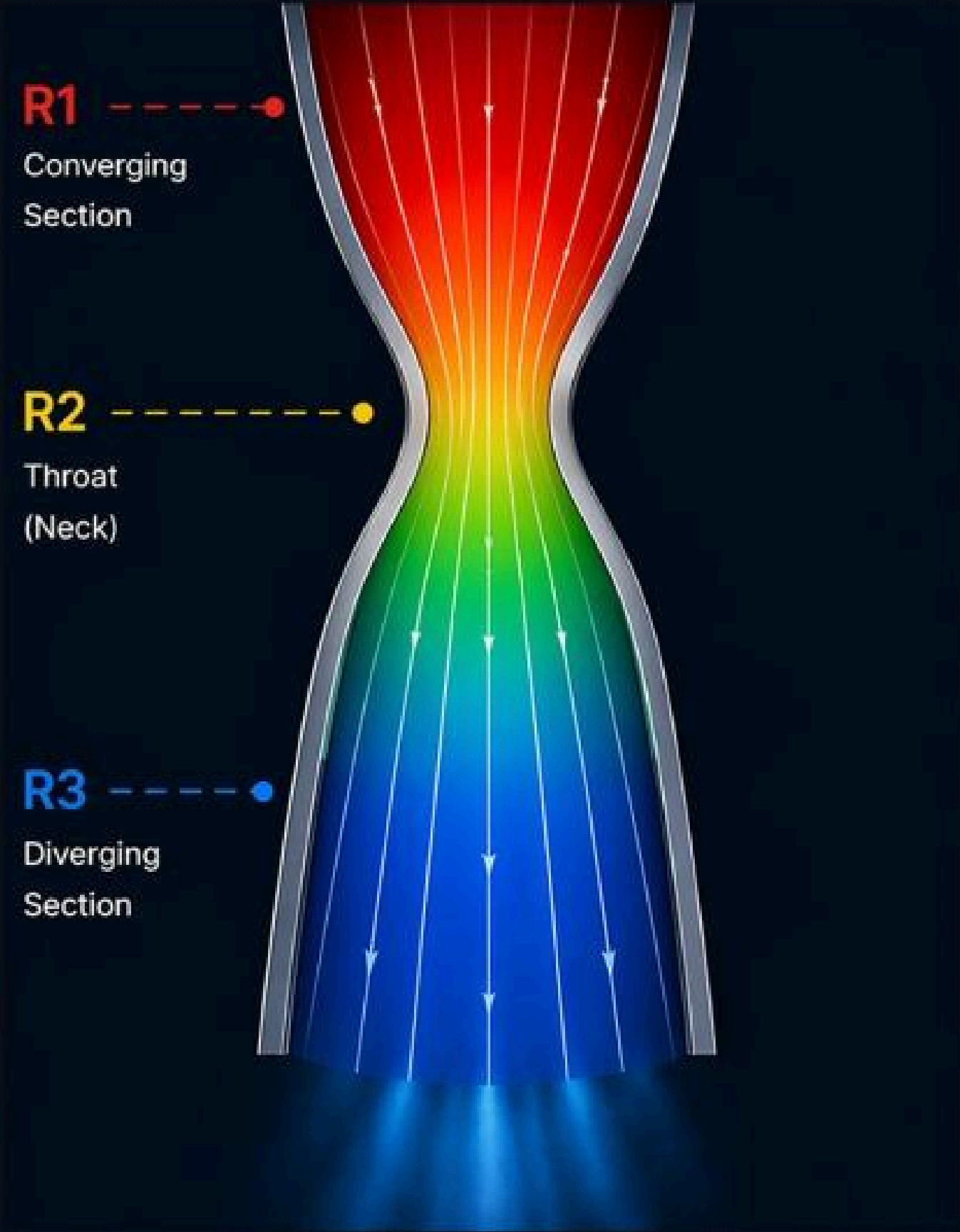
Visualization Mode

- Temperature
- Pressure
- Velocity
- Density

Scope

- Macro
- Micro

Initial: Velocity | Region: R2



Simulation Setup

Select initial conditions to start visualization

Initial Variable

Temperature

Pressure

Velocity

Density

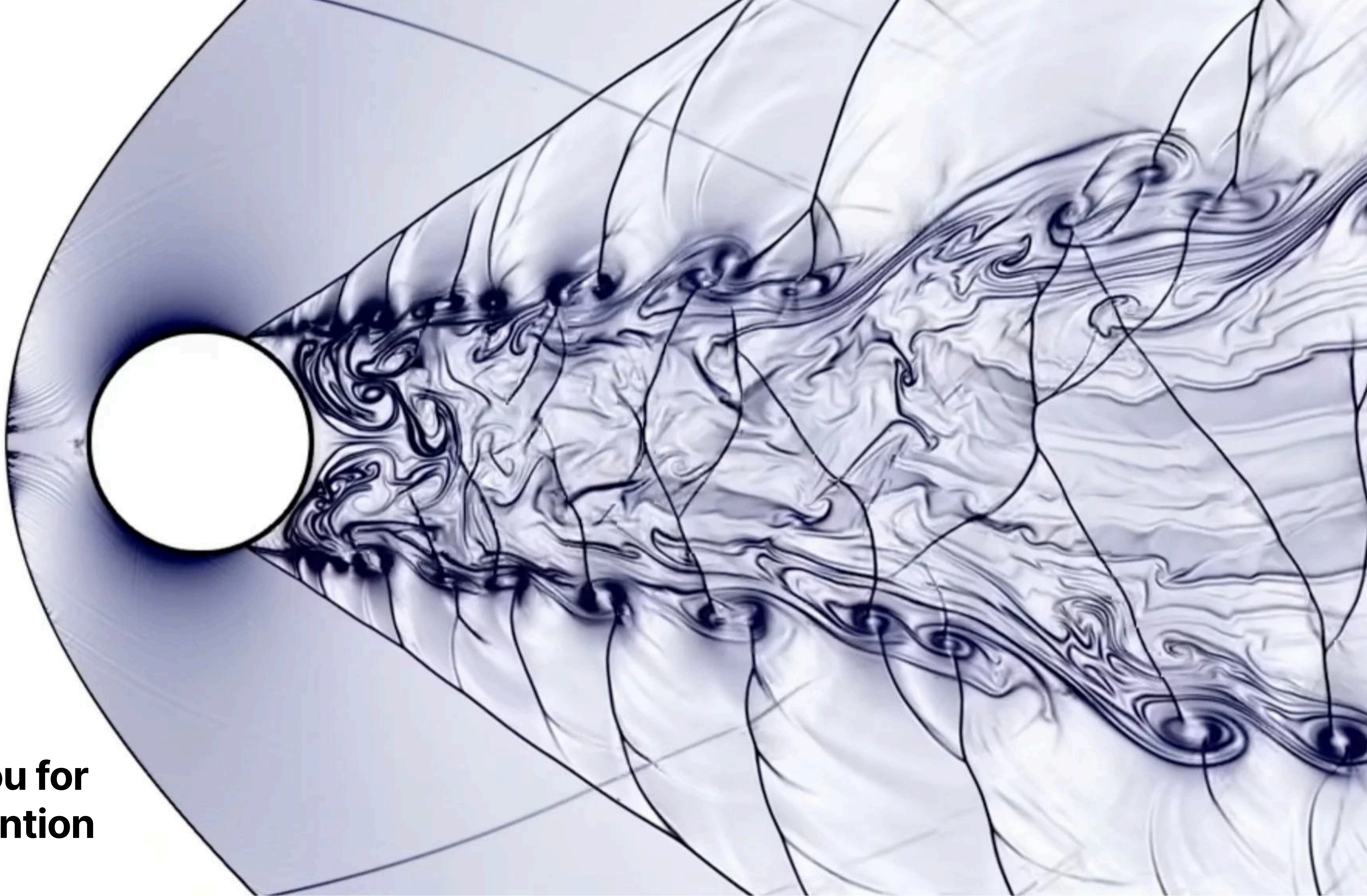
Region

R1

R2

R3

Start



**Thank you for
your attention**